

INTERNAL PRODUCT DOCUMENT

eSlate Platform Process Maps & Feature Inventory

Persona Workflows · Built vs. Needed · Sprint Roadmap

DOCUMENT TYPE

Process Maps + Gap Analysis

BRANCH

trupti-dev

DATE

June 2026

VERSION

1.0

 Admin

 Tutoring Business

 Tutor

 Student

 Parent

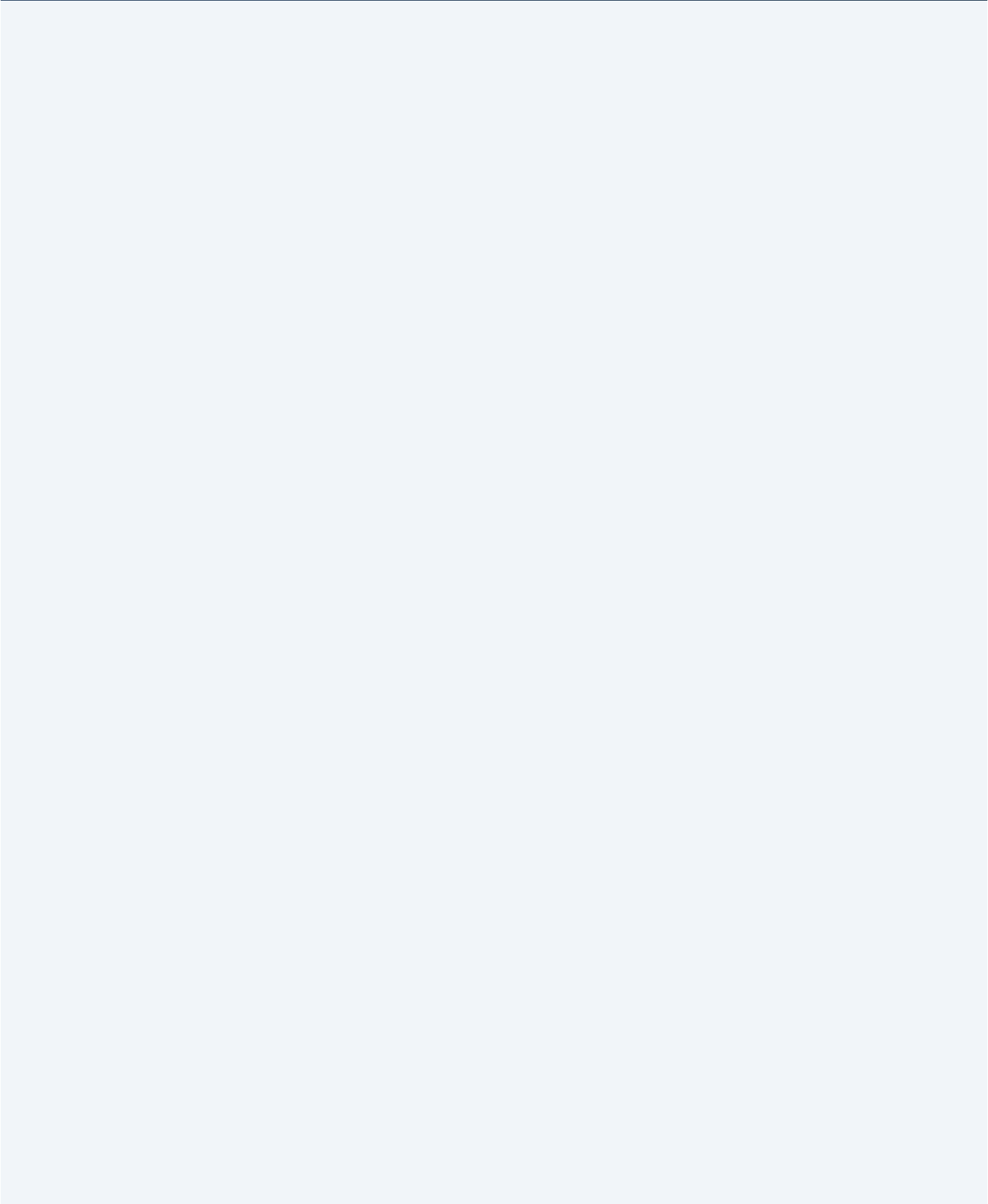


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Process maps, built features, gaps and build roadmap for all 5 personas on the eSlate platform.

PERSONA SECTIONS

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SUMMARY & ROADMAP

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LEGEND

✓ Built & working end-to-end ~ Partially built / stub ✗ Not yet built

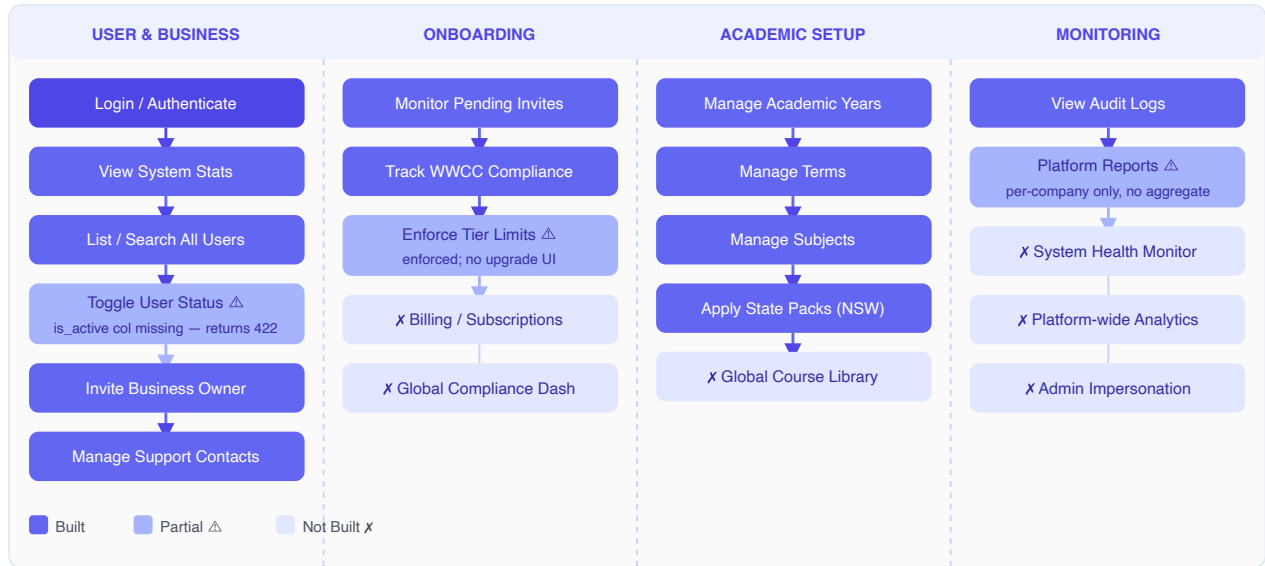
 **Architecture:** eSlate has two backends — **Express/Node.js** (PostgreSQL, port 3000) for the React frontend, and a **Laravel PHP API** (SQLite, Herd at eslate-api.test). Both are covered here.



Admin — Platform Administrator

Role: admin · Single superuser (admin@eslate.com) · Full platform access

Admin Process Map — 4 Core Domains





Admin — Features Built vs. Needed

✓ BUILT & WORKING

- ✓ Login + JWT auth with rate-limit lockout (login_attempts table)
- ✓ Dashboard — live user/business/class/invite counts
- ✓ List all users with role + company filter
- ✓ Invite multi-tutor business owner — creates business + token + email
- ✓ Academic years, terms, subjects CRUD (Laravel)
- ✓ NSW State Packs — curriculum data (GET + apply)
- ✓ View audit logs per company / user
- ✓ Tier enforcement: individual→1, starter→5, pro→20, enterprise→∞
- ✓ Worksheet management + editor
- ✓ Support contacts management (add / remove / list)
- ✓ Company settings CRUD
- ✓ WWCC compliance visibility via tutor records

✗ NEEDS TO BE BUILT

- ✗ **User status toggle** — is_active col missing; endpoint returns 422. Critical fix.
- ✗ Billing & subscription management — no Stripe or payment integration
- ✗ Global compliance dashboard — all tutors' WWCC across all businesses
- ✗ Platform-wide analytics — cross-business trends, DAU/MAU
- ✗ System health monitoring — API errors, DB pool, uptime
- ✗ Global broadcast notifications to all businesses/tutors
- ✗ Global course template library (platform-owned)
- ✗ Admin impersonation — log in as any user for support
- ✗ Tier upgrade/downgrade UI — currently manual
- ✗ Bulk user import via CSV
- ~ Platform reports — per-company only; no aggregate cross-business view

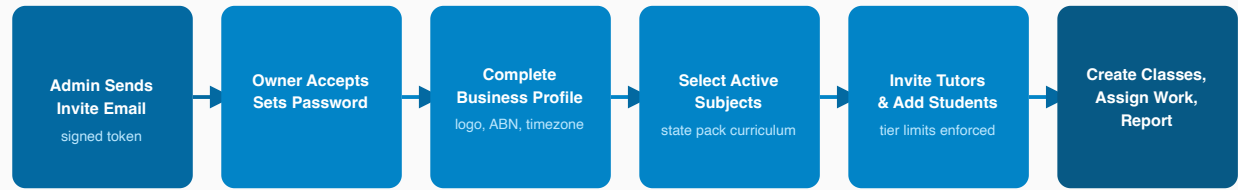
⚠ **Critical fix:** PATCH /api/admin/users/{user}/status fails 422 — is_active column missing from PostgreSQL users table. One migration, 30-minute fix.



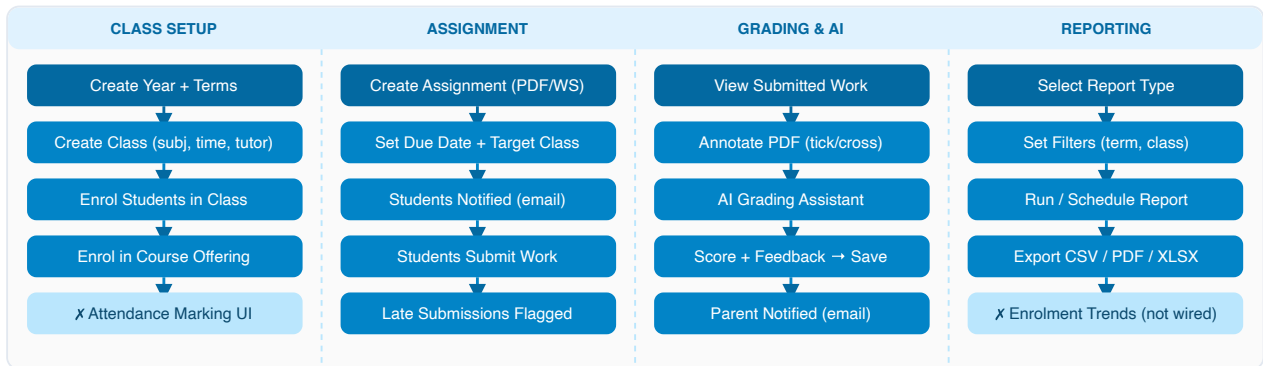
Tutoring Business / Company Admin

Role: company_admin · Manages staff, students, classes, assignments & reporting

Onboarding Flow



Ongoing Operations — Class → Assignment → Grading → Reporting





Tutoring Business — Features Built vs. Needed

✓ BUILT & WORKING

- ✓ Token-based business onboarding (invite → accept → profile)
- ✓ Business profile — logo, ABN, timezone, currency
- ✓ Active subject management (toggle from master list)
- ✓ Academic years + terms CRUD
- ✓ Classes — create, schedule, assign tutor, capacity, conflict detection
- ✓ Enrol / unenrol students in classes
- ✓ Course catalogue — courses + offerings (year × subject × level)
- ✓ Enrol / withdraw students from course offerings
- ✓ Invite tutors — placeholder user + tutor row + email
- ✓ WWCC compliance tracking per tutor (number, expiry, state, status)
- ✓ Add students (record-only, no login required initially)
- ✓ Assignment creation — file-upload or worksheet type
- ✓ File upload to GCS (multer, max 30 MB, PDF/img/doc)
- ✓ View all submissions (by class or business-wide)
- ✓ PDF marking — pen annotation, tick/cross, page rotation
- ✓ AI grading: Gemini → Groq → GPT-4o fallback chain
- ✓ Score + written feedback + reviewer annotations per submission
- ✓ Parent email on graded submission
- ✓ 6 report types + ad-hoc + scheduled runs + CSV/PDF/XLSX export
- ✓ Audit log (all key business actions tracked)
- ✓ Tier enforcement + weekly timetable view

✗ NEEDS TO BE BUILT

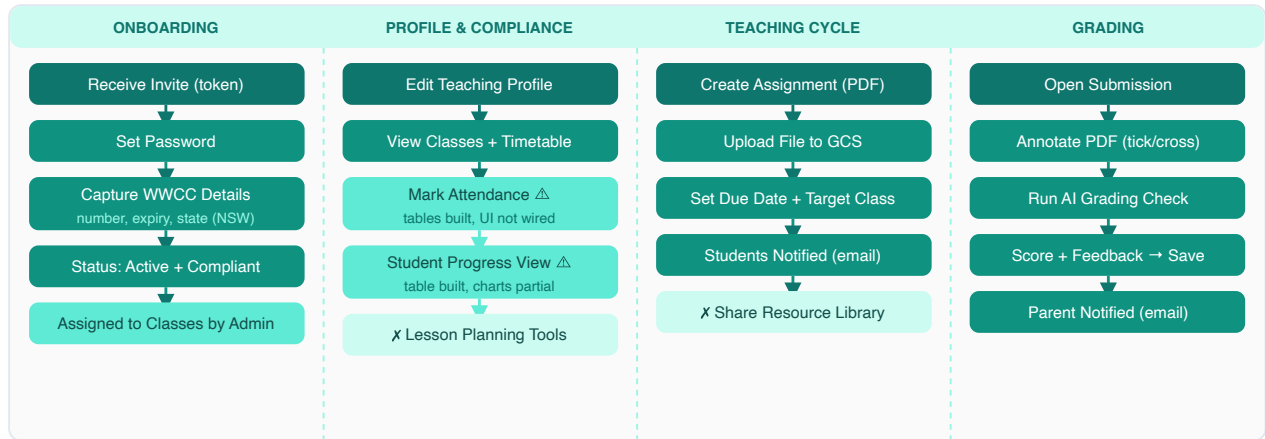
- ✗ Billing & payment integration — no Stripe or invoice generation
- ✗ Real-time messaging — no WebSocket; polling only
- ✗ Calendar sync — no Google Calendar / Outlook integration
- ✗ Automated student progress PDF per student per term
- ✗ Resource / material library — shared file store per business
- ✗ Mobile app (iOS / Android)
- ✗ Live online class / video integration (Zoom / Meet)
- ✗ Tutor payroll / invoicing — hours → pay calculations
- ✗ Enrolment trends report — schema exists, runner not wired
- ✗ Parent self-registration / invite flow
- ✗ Push notifications (browser / mobile)
- ✗ Bulk student import via CSV
- ✗ Multi-branch management UI (branch field on tutors exists)
- ~ Attendance `classSessions` + `sessionAttendance` tables
marking UI
—
built;
front-
end
not
wired
- ~ Holiday / calendar events — tables built; admin form incomplete
- ~ Notification preferences — schema built; UI trigger coverage partial



Tutor

Role: tutor · WWCC compliance required · Assigned to classes by company admin

Tutor Lifecycle & Teaching Process



✓ BUILT

- ✓ Accept invite → set password → capture WWCC → status: compliant
- ✓ 30-day WWCC expiry warning
- ✓ Self-edit profile (bio, rate, delivery modes, year levels)
- ✓ View assigned classes + timetable
- ✓ Create / update / delete assignments (PDF upload)
- ✓ PDF marking: pen strokes, tick/cross, page rotation
- ✓ AI grading (Gemini → Groq → GPT-4o) with reviewer annotations
- ✓ Parent email notification on graded submission

X NEEDS TO BE BUILT

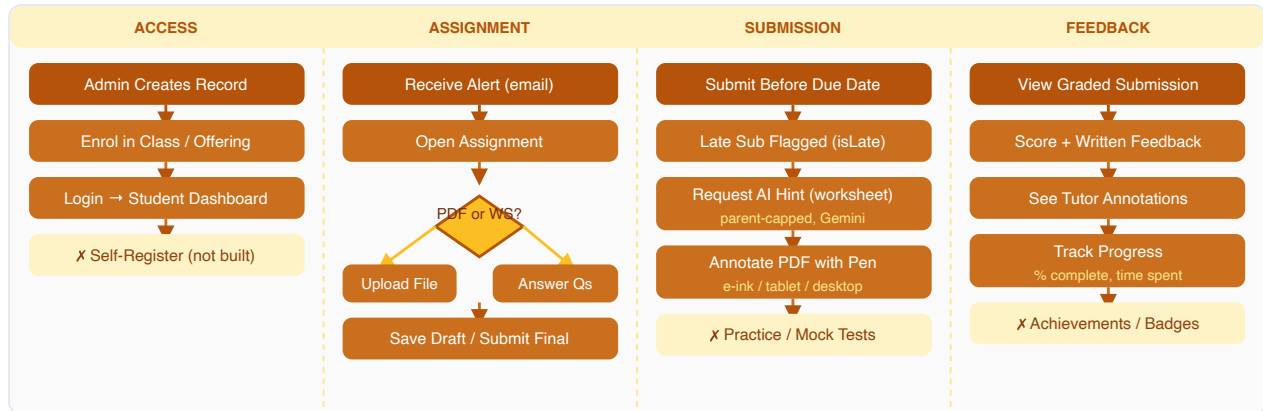
- X Lesson planning tools — no planner or curriculum map
- X Resource / material library — can't share files with students
- X Direct parent messaging channel
- X Tutor invoicing / pay tracking
- X Makeup class / calendar management
- X Assignment templates (save & reuse)
- X Video / live class integration
- ~ Attendance marking UI — DB tables exist; form not built
- ~ Student progress charts — table built; UI partial
- ~ Worksheet builder — schema done; tutor UI incomplete



Student

Role: student · Enrolled by admin · Submits assignments + worksheets · Receives AI hints

Student Learning Process Map



✓ BUILT

- ✓ Dashboard — classes, assignments, messages
- ✓ View assignments (solutions hidden from student)
- ✓ File-upload submission (PDF/img/doc, max 30 MB)
- ✓ Draft saving (`isDraft=true`) until ready to submit
- ✓ Late submission detection + flagging
- ✓ Interactive worksheet — MCQ, fill-blank, text, handwriting
- ✓ PDF pen annotation (e-ink / tablet / desktop) + page rotation
- ✓ AI hints on worksheet questions (parent-capped, Gemini)
- ✓ View score + feedback + tutor annotations on returned work
- ✓ Progress tracking (completion %, time spent)

✗ NEEDS TO BE BUILT

- ✗ Student self-registration — admin must create record; no self-signup
- ✗ Practice / self-study mode outside assigned work
- ✗ Mock exam mode — timed, shuffled, auto-grade MCQ
- ✗ Achievements / badges / streaks (gamification)
- ✗ Study planner — manage own deadlines
- ✗ Push notifications (browser/SMS) — email only currently
- ✗ Offline / PWA mode
- ✗ Parent grade sign-off `parent_verified` DB field, no UI ()
- ✗ Re-submission after `needs_revision` status revision () (incomplete)
- ~ Test / exam UI — full schema built; attempt flow incomplete
- ~ Real-time messaging — MessageCenter built; no WebSocket



Parent Engagement Process Map





Gap Analysis — All Personas

Completion status across all personas and feature categories

PERSONA	FEATURE CATEGORY	STATUS	DONE	KEY GAPS
 Admin	User & Business Mgmt	Partial	75%	User status toggle broken — is_active col missing
	Onboarding & Compliance	Partial	60%	Global compliance dashboard absent; billing completely missing
	Academic Setup	Done	85%	Global course template library not built
	Monitoring & Reports	Gap	40%	No system health monitoring, no platform-wide analytics
	Security & RBAC	Done	90%	No admin impersonation feature
 Business	Onboarding & Profile	Done	90%	Parent invite not linked to business onboarding
	Class & Course Mgmt	Done	82%	Attendance UI not wired; calendar sync absent
	Assignment & Grading	Done	85%	Bulk marking absent; re-submission workflow incomplete
	Staff Management	Partial	70%	Tutor payroll absent; branch management UI not built
	Billing & Communication	Gap	20%	No billing, no real-time messaging, no mobile app
 Tutor	Onboarding & Compliance	Done	95%	Complete — WWCC, profile, status lifecycle all working
	Assignment & Marking	Done	83%	Worksheet builder UI incomplete; bulk marking absent
	Planning & Resources	Gap	8%	No lesson planner, resource library or scheduling tools
 Student	Assignment & Submission	Done	88%	Re-submission + parent sign-off UI missing
	Interactive Learning	Partial	58%	Test/exam UI incomplete; mock exam mode not built
	Engagement & Self-Study	Gap	10%	No practice mode, no push notifications, no self-registration
 Parent	Monitoring & Visibility	Partial	52%	No progress trend charts, no attendance view
	AI Controls	Done	78%	Grade sign-off UI missing; hint tracking complete
	Communication & Billing	Gap	15%	No parent-tutor DM, no billing, no SMS/push, no self-registration



Priority Matrix & Sprint Roadmap

Ranked by impact vs. effort — what to build next for maximum value

🚀 Quick Wins — High Impact, Low Effort

- Fix user status toggle — add `is_active` col to users table (30 min)
- Student self-registration — reuse existing auth + invite route
- Parent self-registration + link-to-child (extend tutor onboarding pattern)
- Attendance marking UI — tables + API exist; just need front-end form
- Re-submission workflow — DB status exists; missing UI state machine
- Parent grade sign-off UI — `parent_verified` field exists; need button
- Calendar events UI — tables fully built; wire to a calendar view
- Real-time messaging — replace polling with WebSocket (socket.io)

📅 Major Features — High Impact, High Effort

- Billing & subscriptions — Stripe, invoices, tier upgrades/downgrades
- Test / exam module (student UI) — attempt flow, timer, auto-grade MCQ
- Tutor lesson planner + resource / material library
- Video / live class integration (Zoom / Meet embed)
- Mobile app (React Native or PWA)
- Automated student progress report PDF per term
- Platform-wide admin analytics dashboard
- Bulk student import via CSV

⚡ Nice-to-Have — Lower Urgency

- Admin impersonation (log in as any user for support)
- Assignment templates (save & reuse)
- Global course template library (platform-owned)
- Tutor invoice / pay stub generation
- Enrolment trends report wiring (schema done)
- Holiday calendar admin form (DB table built)
- Parent-teacher conference booking flow

🕒 Future / Deferred

- SMS notifications (Twilio) — email working; SMS nice-to-have
- Student gamification — badges, streaks, leaderboards
- Offline / PWA mode for students
- Google Calendar / Outlook sync
- Multi-branch management UI
- AI question generator for tutors
- System health monitoring dashboard

Suggested Sprint Order

Sprint 1 — Fix & Unlock

- Fix user status toggle
- Student self-register
- Parent self-register
- Attendance marking UI

Sprint 2 — Engagement

- Real-time messaging (WS)
- Parent grade sign-off UI
- Re-submission workflow
- Calendar events UI

Sprint 3 — Learning

- Test / exam module
- Progress trend charts
- Worksheet builder polish
- Tutor resource library

Sprint 4+ — Scale

- Billing integration
- Auto progress reports
- Video / live class
- Mobile / PWA

